What Lovelock proposed was not without precedent. In the 1920s, Vladimir Vernadsky, a Russian scientist, had developed the idea of the biosphere, the zone of Earth that holds all living organisms, and suggested that it could be seen as a single entity in which organic and inorganic elements interact. The British botanist Arthur Tansley then took this idea further in the 1930s, with his concept of an “ecosystem” that regulates itself into a state of equilibrium.

Tansley’s theory was at the heart of Lovelock’s hypothesis: that all living organisms and their environment form one complex super-ecosystem that regulates and balances conditions to sustain life on Earth. The idea first occurred to Lovelock in the late 1960s, but it was after discussing it with US microbiologist Lynn Margulis that it began to take shape. Together, they presented the hypothesis in a paper in 1974, giving it a name suggested by the writer William Golding—Gaia, after the ancient Greek Earth goddess. Lovelock and Margulis portrayed Earth as a living entity, composed of the biosphere, living organisms; the pedosphere, the surface layer of the Earth; the hydrosphere, the bodies of water on the Earth’s surface; and the atmosphere, the gases surrounding the Earth. These spheres and their complex interactions maintain Earth in “homeostasis.” This concept is borrowed from physiology, which describes the stable internal conditions, such as temperature and chemical composition, that allow organisms to function optimally. They are controlled by self-regulating mechanisms that react to change in those conditions. Lovelock’s use of the word homeostasis reinforced the implication that Earth, or Gaia, is a living entity.